Cambridge Assessment
International Education

Cambridge IGCSE™

CANDIDATE NAME			
CENTRE NUMBER		CANDIDATE NUMBER	

PHYSICS **0625/42**

Paper 4 Theory (Extended) **Practice Paper**

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].

This document has 8 pages. Any blank pages are indicated.

1 Which instrument is most suitable to determine the volume of a small irregularly shaped stone?

A 30 cm ruler

B digital timer

C measuring cylinder

D tape measure

2 A steel ball is dropped from the top floor of a building. Air resistance can be ignored.

Which statement describes the motion of the ball?

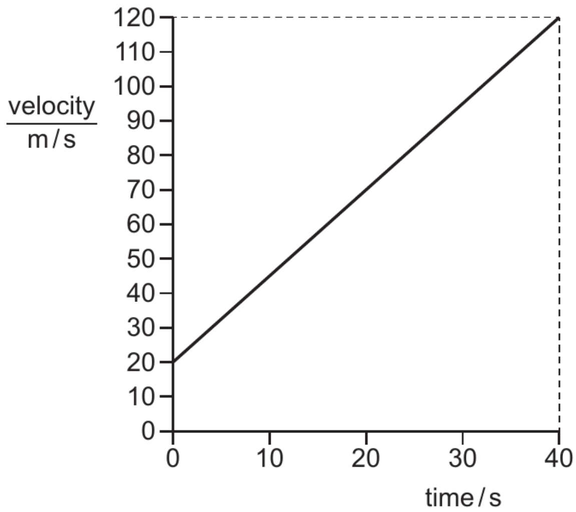
A The ball falls with constant acceleration.

B The ball falls with constant speed.

C The ball falls with decreasing speed.

D The ball falls with increasing acceleration.

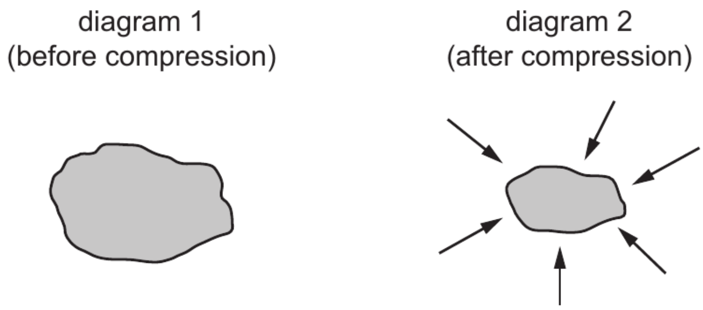
3 The diagram shows a velocity-time graph for an object which is accelerating.



What is the acceleration of the object?

A 0.40 m/s^2 **B** 2.5 m/s^2 **C** 3.0 m/s^2 **D** 100 m/s^2

4 Diagram 1 shows a piece of flexible material that contains many pockets of air. Diagram 2 shows the same piece of flexible material after it has been compressed so that its volume decreases.



What happens to the mass and to the weight of the flexible material when it is compressed?

	mass	weight
A	increases	increases
B	increases	no change
C	no change	increases
D	no change	no change

5 A student determines the density of copper. She has a rectangular block of copper.

Which other apparatus does she need?

	balance	ruler	stop-watch	thermometer
A	✓	✓	✓	✓
B	✓	✓	✓	*x*
C	✓	✓	*x*	*x*
D	✓	*x*	*x*	*x*

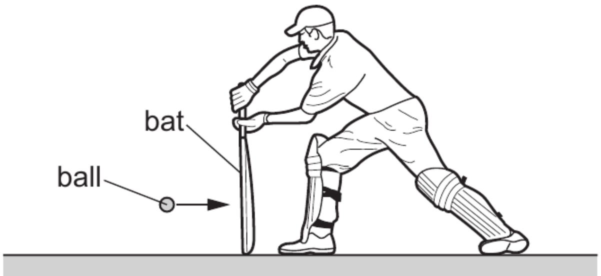
6 A 0.15 kg mass is suspended from a vertical spring. The mass is pulled downwards until the spring has extended by 12 cm from its unstretched length.

The spring constant of the spring is 30 N/m. What is the magnitude of the initial acceleration of the spring when it is released?

A $14\ m/s^2$ **B** $24\ m/s^2$ **C** $28\ m/s^2$ **D** $34\ m/s^2$

7 A cricket ball of mass 0.15 kg is moving at a speed of 30 m/s.

The diagram shows a batsman holding a bat stationary as the ball hits the bat and bounces back at the same speed in the opposite direction.



The ball is in contact with the bat for 1.0×10^{-3} s. Which row is correct?

	change of momentum kg m/s	force applied to ball / N
A	4.5	4500
B	4.5	9000
C	9.0	4500
D	9.0	9000

8 A stone is thrown vertically upwards at a speed of 5.0 m/s.

All of the energy initially in the kinetic store of the stone is transferred to its gravitational potential store when the stone reaches its maximum height. Which equation gives the maximum height h reached by the stone?

A $h = \dfrac{9.8}{2 \times 5.0^2}$

B $h = \dfrac{5.0^2 \times 2}{9.8}$

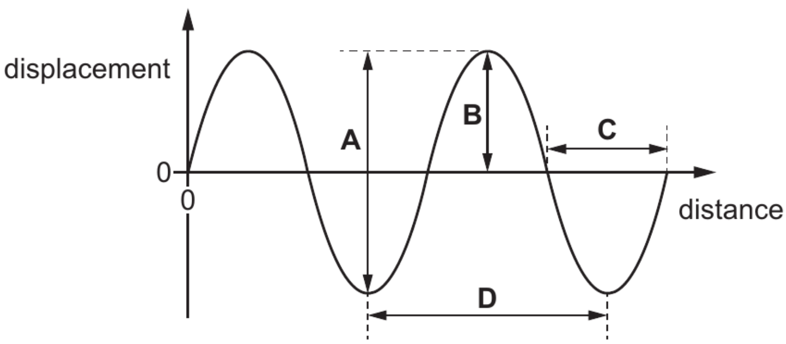
C $h = \dfrac{5.0^2}{2 \times 9.8}$

D $h = \dfrac{2 \times 9.8}{5.0^2}$

9 Which wave will diffract the most when passing through a gap that is 0.85 m wide?

	wave speed m/s	wave frequency / Hz
A	300	280
B	300	400
C	350	280
D	350	400

10 Which arrow on the graph shows the amplitude of the wave?

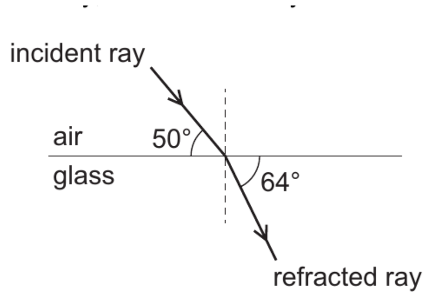


11 Which row describes a seismic S-wave?

	type of wave	direction of vibration
A	longitudinal	parallel to the direction of travel of the wavefront
B	longitudinal	perpendicular to the direction of travel of the wavefront
C	transverse	parallel to the direction of travel of the wavefront
D	transverse	perpendicular to the direction of travel of the wavefront

12 A ray of light, travelling in air, is incident on an air-glass boundary.

The angles between the incident ray, the refracted ray and the boundary are shown.



to calculate the refractive index *n* of the glass

Which equation is used to calculate the refractive index n of the glass?

A $n = \dfrac{\sin 40°}{\sin 26°}$ **B** $n = \dfrac{\sin 40°}{\sin 64°}$ **C** $n = \dfrac{\sin 50°}{\sin 26°}$ **D** $n = \dfrac{\sin 50°}{\sin 64°}$

13 A small object O is placed near a converging lens, as shown. The lens forms an image I.

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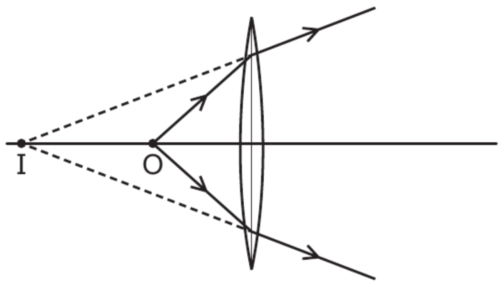


Which statement is correct?

A The image I is diminished.

B The image I is inverted.

C The image I is real.

D The object O is closer to the lens than its principal focus.

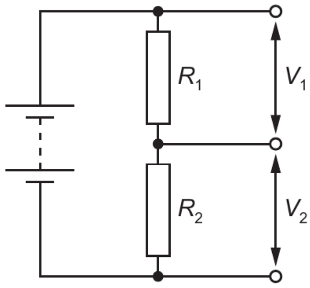
14 A metallic wire X has a resistance.

What is the difference in resistance for a wire of the same material as X that is shorter than X but the same diameter, and for a wire that has the same length as X but a smaller diameter?

	resistance of shorter wire	resistance of wire with smaller diameter
A	greater	greater
B	greater	smaller
C	smaller	greater
D	smaller	smaller

15 The diagram shows the circuit for a potential divider.

ıit for a potential divider.



Which equation is used to calculate V_1?

A $V_1 = \dfrac{R_1}{V_2 \times (R_1 + R_2)}$

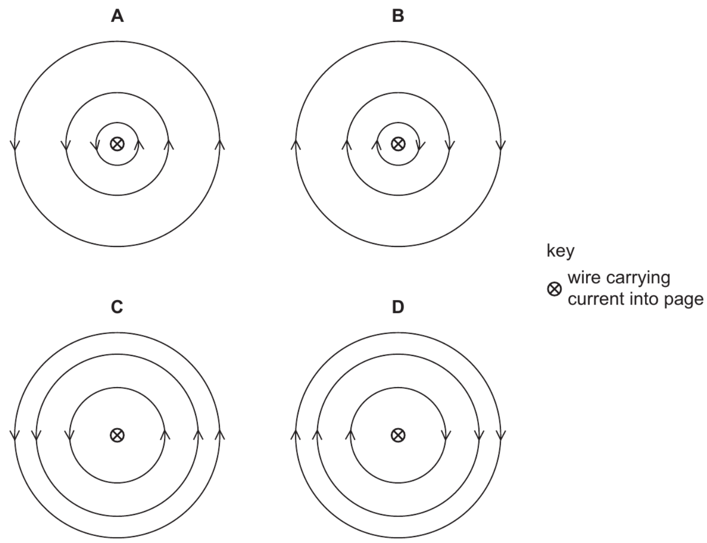
B $V_1 = \dfrac{V_2 \times R_1}{R_2}$

C $V_1 = \dfrac{V_2 \times R_1}{R_1 + R_2}$

D $V_1 = \dfrac{R_1}{V_2 \times R_2}$

16 When there is an electric current in a long straight wire, a magnetic field is created around the wire.

Which diagram shows the correct pattern and direction of magnetic field lines around a long straight wire carrying current into the page?



17 When a thin gold foil is bombarded with alpha particles, some alpha particles are deflected through large angles.

Which statement explains this deflection?

A Most of the atom consists of empty space.

B All of the positive charge and most of the mass of the gold atom are concentrated in a small volume.

C Positive charge in the gold atom is spread evenly throughout the atom.

D All of the negative charge is concentrated at its centre.